

Profiling the Scanpy PBMC Workflow

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1. Profiling methods

I ran the supplied Scanpy workflow on the PBMC3K, PBMC6K, and PBMC10K datasets on an OddJobs compute node. Each Slurm job requested one CPU, 16 GiB of memory, and a two-hour limit in the overflow partition. The environment used Python 3.12.12, Scanpy 1.10.2, and AnnData 0.10.8. OpenMP, OpenBLAS, MKL, and Numba were limited to one thread. Jobs 1290633, 1290634, and 1290635 completed with exit code 0:0.

Whole-process measurements used `/usr/bin/time -v`. I measured ten workflow sections with `time.perf_counter()` and used cProfile around `sc.tl.rank_genes_groups` with the Wilcoxon method. Each dataset was run once with each profiling method. The plots use the nominal dataset sizes of 3,000, 6,000, and 10,000 cells, rather than the number remaining after filtering.

2. Coarse-grain results

Dataset	Elapsed (s)	CPU	User (s)	System (s)	Max RSS (MiB)
PBMC3K	18.48	96%	17.09	0.66	835.1
PBMC6K	26.05	97%	24.43	0.94	890.9
PBMC10K	59.44	99%	54.53	4.38	1841.7

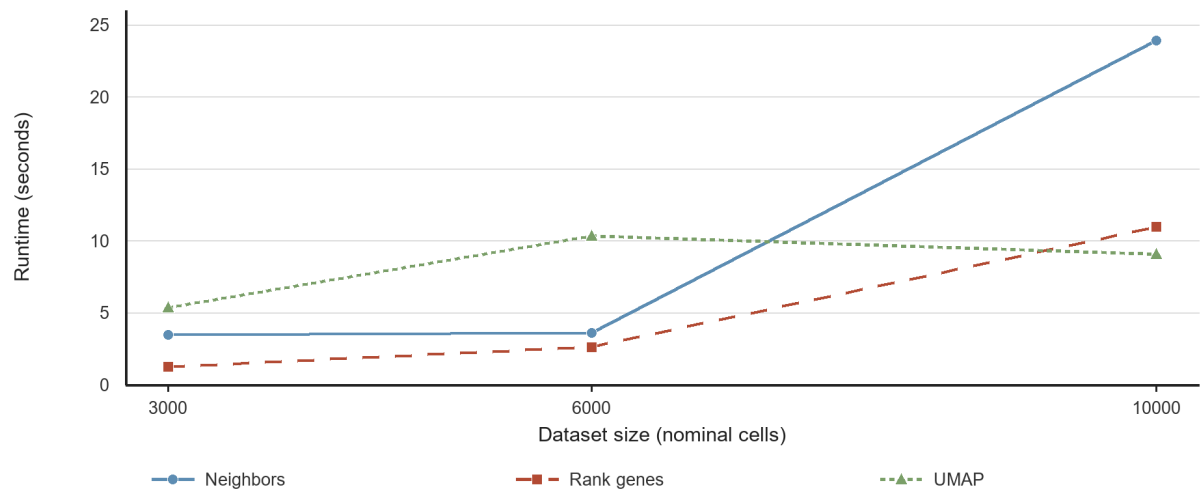
Table 1. Whole-process measurements. Bold columns highlight elapsed time, CPU utilization, and peak resident memory. Memory values were converted from KiB to MiB by dividing by 1,024.

Elapsed time increased from 18.48 seconds at 3K to 59.44 seconds at 10K. Peak memory increased from 835.1 to 1841.7 MiB. CPU utilization stayed close to 100%, which is consistent with a workload using roughly one CPU over most of the run.

The console output shows that the 3K coarse run read an existing cache, whereas the 6K and 10K coarse runs wrote a cache. The subsequent instrumented and cProfile runs read cached input for all three datasets. The timings therefore describe these recorded runs; they are not controlled measurements of cold-start input loading. Full console output and GNU time records appear in Section 7.

3. Instrumented profiling

Longer sections



Shorter sections

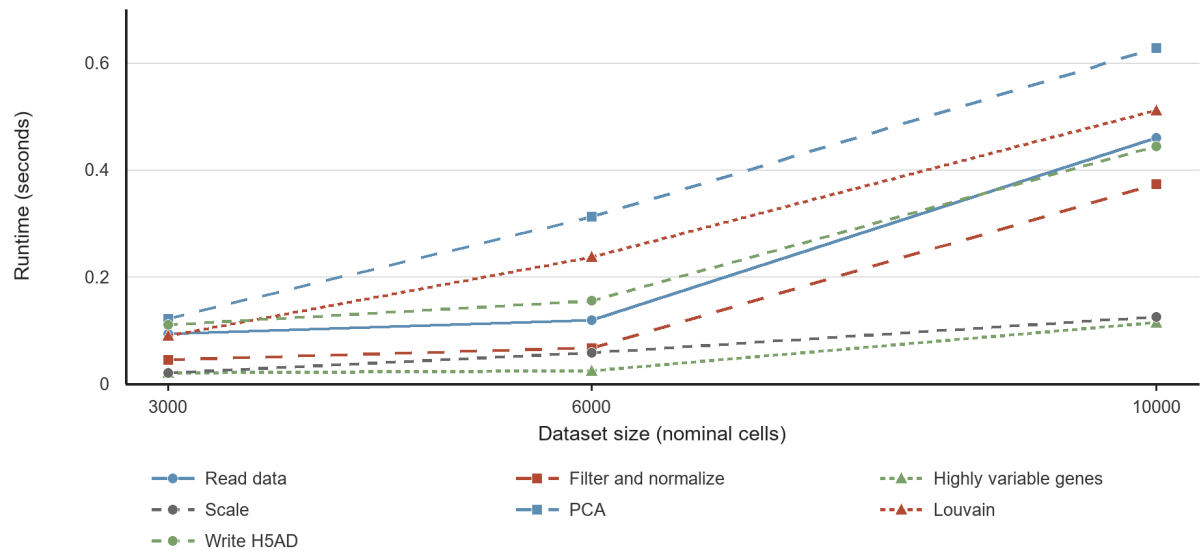


Figure 1. Section runtime at the three nominal dataset sizes. The two panels use different vertical scales so that short sections remain visible. Points are individual measurements; connecting lines show the observed changes.

At 10K, **neighbors** was the largest measured section at **23.915 seconds**, followed by gene ranking at 10.944 seconds and UMAP at 9.058 seconds. At 3K and 6K, UMAP was the largest section. The bottleneck therefore changed across the tested datasets.

4. Fine-grain profiling

PBMC3K

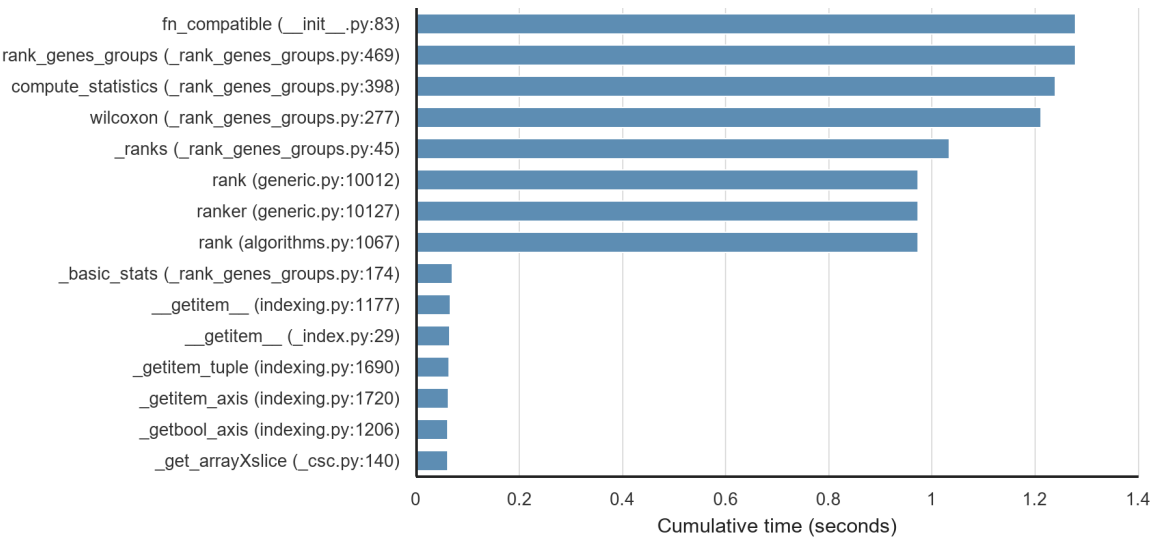


Figure 2. The 15 functions with the largest cumulative times in the PBMC3K gene-ranking profile. Labels give function names and source locations.

PBMC6K

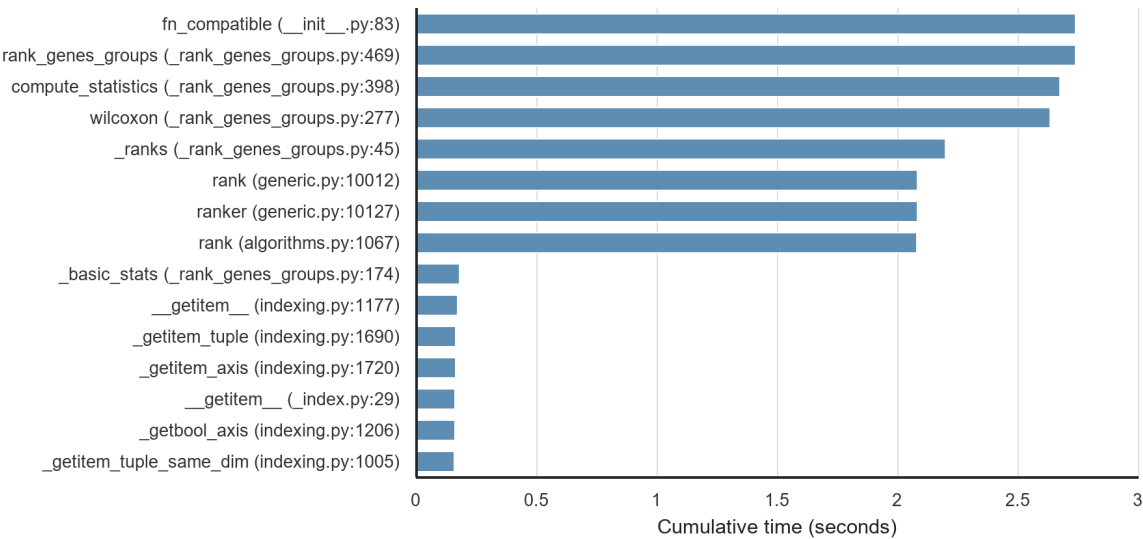


Figure 3. The same ranking for PBMC6K. Cumulative times include calls made by each function, so the bars overlap in what they measure and must not be summed.

4. Fine-grain profiling (continued)

PBMC10K

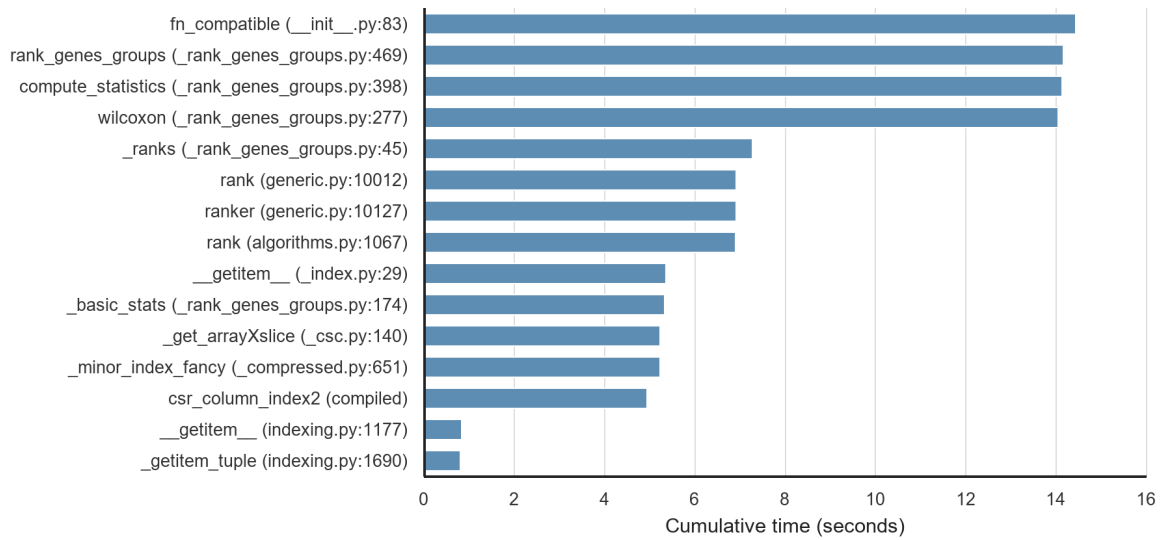


Figure 4. The 15 functions with the largest cumulative times in the PBMC10K gene-ranking profile. The horizontal scale differs from Figures 2 and 3 to show each dataset clearly.

The Wilcoxon calculation and its rank computation account for most of the cumulative time within gene ranking. The pandas ranking routine in `algorithms.py` took about 0.974 seconds at 3K, 2.083 seconds at 6K, and 6.918 seconds at 10K. In the 10K profile, `rank_genes_groups` took 14.175 seconds. Its basic-statistics path took 5.334 seconds, and sparse CSC slicing took about 5.246 seconds. These nested times describe overlapping work.

The cProfile measurements were collected in separate runs and include profiling overhead. They should not be substituted for the section timings. They explain work inside gene ranking but do not profile the neighbor-graph step, which was the largest section in the 10K instrumented run.

5. Bottlenecks and expected runtime at 20K

Neighbors showed the largest absolute increase, rising by 20.464 seconds between 3K and 10K. Gene ranking rose by 9.710 seconds, or about 8.9 times its 3K runtime. This was the largest proportional increase among the three dominant sections. Filter and normalize rose by about 8.3 times, but still took less than 0.4 seconds at 10K. UMAP did not increase monotonically: its measured runtime was lower at 10K than at 6K.

I fitted a separate least-squares line to the three recorded times for each section and evaluated it at 20,000 nominal cells. Table 2 reports these extrapolations alongside the measurements. The fitted neighbor step is the largest predicted cost, followed by gene ranking.

Section	3K (s)	6K (s)	10K (s)	Predicted 20K (s)
Read data	0.093	0.118	0.460	0.965
Filter and normalize	0.045	0.067	0.373	0.825
Highly variable genes	0.020	0.024	0.114	0.244
Scale	0.020	0.058	0.125	0.273
PCA	0.122	0.312	0.627	1.346
Neighbors	3.451	3.601	23.915	51.869
Louvain	0.089	0.237	0.511	1.109
UMAP	5.348	10.326	9.058	14.861
Write H5AD	0.110	0.155	0.444	0.907
Rank genes	1.234	2.620	10.944	24.403

Table 2. Measured section times and linear extrapolations. Bold values identify neighbors, the largest section in the 10K run and the largest fitted cost at 20K.

The section predictions sum to 96.80 seconds. This sum excludes startup, imports, and any work outside the timed sections, so it is not a prediction of the full process measured by GNU time. Only three dataset sizes and one timing per method and dataset were available. Differences in data composition, algorithm behavior, caching, and run-to-run variability may affect the fitted trends. In particular, the UMAP measurements do not support a reliable linear scaling claim. Repeated runs and a measured 20K dataset would be needed to assess prediction accuracy.

6. Slurm batch scripts

The three scripts below reproduce the experiment sequence. The outputs directory must exist before submitting a job because Slurm opens its log files before the script runs.

PBMC3K: slurm/pbmc3k.sh

```
#!/bin/bash -l
#SBATCH -J pbmc3k
#SBATCH -p overflow
#SBATCH --mem=16G
#SBATCH --cpus-per-task=1
#SBATCH --time=2:00:00
#SBATCH --output=outputs/pbmc3k_slurm_%j.out
#SBATCH --error=outputs/pbmc3k_slurm_%j.err

set -euo pipefail
cd "$SLURM_SUBMIT_DIR"

module load bmi/python-3.12.12
source "$HOME/venvs/bmi500-py312/bin/activate"

export OMP_NUM_THREADS=1
export OPENBLAS_NUM_THREADS=1
export MKL_NUM_THREADS=1
export NUMBA_NUM_THREADS=1

mkdir -p outputs results

/usr/bin/time -v \
  -o outputs/pbmc3k_coarse_time.txt \
  python scanpy_pbmc.py \
    --data-dir data \
    --data-set pbmc3k \
    --out-dir results \
    --num-threads 1 \
  > outputs/pbmc3k_coarse_console.txt 2>&1

python scanpy_pbmc_instrumented.py \
  --data-dir data \
  --data-set pbmc3k \
  --out-dir results \
  --num-threads 1 \
  > outputs/pbmc3k_instrumented_console.txt 2>&1

python scanpy_pbmc_cprofile.py \
  --data-dir data \
  --data-set pbmc3k \
  --out-dir results \
  --num-threads 1 \
  > outputs/pbmc3k_cprofile_console.txt 2>&1
```

6. Slurm batch scripts (continued)

PBMC6K: slurm/pbmc6k.sh

```
#!/bin/bash -l
#SBATCH -J pbmc6k
#SBATCH -p overflow
#SBATCH --mem=16G
#SBATCH --cpus-per-task=1
#SBATCH --time=2:00:00
#SBATCH --output=outputs/pbmc6k_slurm_%j.out
#SBATCH --error=outputs/pbmc6k_slurm_%j.err

set -euo pipefail
cd "$SLURM_SUBMIT_DIR"

module load bmi/python-3.12.12
source "$HOME/venvs/bmi500-py312/bin/activate"

export OMP_NUM_THREADS=1
export OPENBLAS_NUM_THREADS=1
export MKL_NUM_THREADS=1
export NUMBA_NUM_THREADS=1

mkdir -p outputs results

/usr/bin/time -v \
-o outputs/pbmc6k_coarse_time.txt \
python scanpy_pbmc.py \
--data-dir data \
--data-set pbmc6k \
--out-dir results \
--num-threads 1 \
> outputs/pbmc6k_coarse_console.txt 2>&1

python scanpy_pbmc_instrumented.py \
--data-dir data \
--data-set pbmc6k \
--out-dir results \
--num-threads 1 \
> outputs/pbmc6k_instrumented_console.txt 2>&1

python scanpy_pbmc_cprofile.py \
--data-dir data \
--data-set pbmc6k \
--out-dir results \
--num-threads 1 \
> outputs/pbmc6k_cprofile_console.txt 2>&1
```

6. Slurm batch scripts (continued)

PBMC10K: slurm/pbmc10k.sh

```
#!/bin/bash -l
#SBATCH -J pbmc10k
#SBATCH -p overflow
#SBATCH --mem=16G
#SBATCH --cpus-per-task=1
#SBATCH --time=2:00:00
#SBATCH --output=outputs/pbmc10k_slurm_%j.out
#SBATCH --error=outputs/pbmc10k_slurm_%j.err

set -euo pipefail
cd "$SLURM_SUBMIT_DIR"

module load bmi/python-3.12.12
source "$HOME/venvs/bmi500-py312/bin/activate"

export OMP_NUM_THREADS=1
export OPENBLAS_NUM_THREADS=1
export MKL_NUM_THREADS=1
export NUMBA_NUM_THREADS=1

mkdir -p outputs results

/usr/bin/time -v \
-o outputs/pbmc10k_coarse_time.txt \
python scanpy_pbmc.py \
--data-dir data \
--data-set pbmc10k \
--out-dir results \
--num-threads 1 \
> outputs/pbmc10k_coarse_console.txt 2>&1

python scanpy_pbmc_instrumented.py \
--data-dir data \
--data-set pbmc10k \
--out-dir results \
--num-threads 1 \
> outputs/pbmc10k_instrumented_console.txt 2>&1

python scanpy_pbmc_cprofile.py \
--data-dir data \
--data-set pbmc10k \
--out-dir results \
--num-threads 1 \
> outputs/pbmc10k_cprofile_console.txt 2>&1
```


7. Console outputs

PBMC3K: coarse-grain run

```
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/louvain/__init__.py:54: UserWarning: pkg_resources is deprecated as an
API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as
2025-11-30. Refrain from using this package or pin to Setuptools<81.
  from pkg_resources import get_distribution, DistributionNotFound
... reading from cache file cache/data-pbmc3k-filtered_gene_bc_matrices-matrix.h5ad
filtered out 19024 genes that are detected in less than 3 cells
normalizing counts per cell
  finished (0:00:00)
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an
AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
  with n_comps=30
  finished (0:00:00)
computing neighbors
  using 'X_pca' with n_pcs = 30
  finished: added to `uns['neighbors']`
  `obsp['distances']`, distances for each pair of neighbors
  `obsp['connectivities']`, weighted adjacency matrix (0:00:03)
running louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 4 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:05)
ranking genes
  finished: added to `uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:01)
scanpy==1.10.2 anndata==0.10.8 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph
==0.11.6 louvain==0.8.2 pynndescent==0.6.0
using 1 threads
```

GNU time output

```
Command being timed: "python scanpy_pbmc.py --data-dir data --data-set pbmc3k --out-dir results --num-threads 1"
User time (seconds): 17.09
System time (seconds): 0.66
Percent of CPU this job got: 96%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:18.48
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 855148
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 110
Minor (reclaiming a frame) page faults: 148780
Voluntary context switches: 10839
Involuntary context switches: 82
Swaps: 0
File system inputs: 0
File system outputs: 86304
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0
```

7. Console outputs (continued)

PBMC6K: coarse-grain run

```
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/louvain/__init__.py:54: UserWarning: pkg_resources is deprecated as an
API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as
2025-11-30. Refrain from using this package or pin to Setuptools<81.
  from pkg_resources import get_distribution, DistributionNotFound
... writing an h5ad cache file to speedup reading next time
filtered out 1 cells that have less than 200 genes expressed
filtered out 18395 genes that are detected in less than 3 cells
normalizing counts per cell
  finished (0:00:00)
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an
AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
  with n_comps=30
  finished (0:00:00)
computing neighbors
  using 'X_pca' with n_pcs = 30
  finished: added to `uns['neighbors']`
  `obsp['distances']`, distances for each pair of neighbors
  `obsp['connectivities']`, weighted adjacency matrix (0:00:03)
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 7 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:10)
ranking genes
  finished: added to `uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:02)
scanpy==1.10.2 anndata==0.10.8 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igrph
==0.11.6 louvain==0.8.2 pynndescent==0.6.0
using 1 threads
```

GNU time output

```
Command being timed: "python scanpy_pbmc.py --data-dir data --data-set pbmc6k --out-dir results --num-threads 1"
User time (seconds): 24.43
System time (seconds): 0.94
Percent of CPU this job got: 97%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:26.05
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 912288
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 134
Minor (reclaiming a frame) page faults: 212949
Voluntary context switches: 12406
Involuntary context switches: 123
Swaps: 0
File system inputs: 99144
File system outputs: 197384
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0
```

7. Console outputs (continued)

PBMC10K: coarse-grain run

```
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/louvain/__init__.py:54: UserWarning: pkg_resources is deprecated as an
  API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as
  2025-11-30. Refrain from using this package or pin to Setuptools<81.
  from pkg_resources import get_distribution, DistributionNotFound
... writing an h5ad cache file to speedup reading next time
filtered out 232 cells that have less than 200 genes expressed
filtered out 13246 genes that are detected in less than 3 cells
normalizing counts per cell
  finished (0:00:00)
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an
  AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
  with n_comps=30
  finished (0:00:00)
computing neighbors
  using 'X_pca' with n_pcs = 30
  finished: added to `.uns['neighbors']`
  `.obsp['distances']`, distances for each pair of neighbors
  `.obsp['connectivities']`, weighted adjacency matrix (0:00:25)
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 16 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:09)
ranking genes
  finished: added to `.uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:11)
scanpy==1.10.2 anndata==0.10.8 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph
==0.11.6 louvain==0.8.2 pynndescent==0.6.0
using 1 threads
```

GNU time output

```
Command being timed: "python scanpy_pbmc.py --data-dir data --data-set pbmc10k --out-dir results --num-threads 1"
User time (seconds): 54.53
System time (seconds): 4.38
Percent of CPU this job got: 99%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:59.44
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 1885920
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 139
Minor (reclaiming a frame) page faults: 930630
Voluntary context switches: 20987
Involuntary context switches: 235
Swaps: 0
File system inputs: 622264
File system outputs: 831272
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0
```

7. Console outputs (continued)

PBMC3K: instrumented run

```
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/louvain/__init__.py:54: UserWarning: pkg_resources is deprecated as an
  API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as
  2025-11-30. Refrain from using this package or pin to Setuptools<81.
  from pkg_resources import get_distribution, DistributionNotFound
... reading from cache file cache/data-pbmc3k-filtered_gene_bc_matrices-matrix.h5ad
filtered out 19024 genes that are detected in less than 3 cells
normalizing counts per cell
  finished (0:00:00)
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an
  AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
  with n_comps=30
  finished (0:00:00)
computing neighbors
  using 'X_pca' with n_pcs = 30
  finished: added to `uns['neighbors']`
  `.obsp['distances']`, distances for each pair of neighbors
  `.obsp['connectivities']`, weighted adjacency matrix (0:00:03)
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 4 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:05)
ranking genes
  finished: added to `uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:01)
scanpy==1.10.2 anndata==0.10.8 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph
==0.11.6 louvain==0.8.2 pyndescent==0.6.0
using 1 threads
wrote section timings to results/pbmc3k_section_times.csv
```

7. Console outputs (continued)

PBMC6K: instrumented run

```
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/louvain/__init__.py:54: UserWarning: pkg_resources is deprecated as an
  API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as
  2025-11-30. Refrain from using this package or pin to Setuptools<81.
from pkg_resources import get_distribution, DistributionNotFound
... reading from cache file cache/data-pbmc6k-filtered_gene_bc_matrices-matrix.h5ad
filtered out 1 cells that have less than 200 genes expressed
filtered out 18395 genes that are detected in less than 3 cells
normalizing counts per cell
  finished (0:00:00)
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an
  AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
  with n_comps=30
  finished (0:00:00)
computing neighbors
  using 'X_pca' with n_pcs = 30
  finished: added to `.uns['neighbors']`
  `.obsp['distances']`, distances for each pair of neighbors
  `.obsp['connectivities']`, weighted adjacency matrix (0:00:03)
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 7 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:10)
ranking genes
  finished: added to `.uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:02)
scanpy==1.10.2 anndata==0.10.8 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph
==0.11.6 louvain==0.8.2 pynndescent==0.6.0
using 1 threads
wrote section timings to results/pbmc6k_section_times.csv
```

7. Console outputs (continued)

PBMC10K: instrumented run

```
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/louvain/__init__.py:54: UserWarning: pkg_resources is deprecated as an
  API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as
  2025-11-30. Refrain from using this package or pin to Setuptools<81.
from pkg_resources import get_distribution, DistributionNotFound
... reading from cache file cache/data-pbmc10k-filtered_gene_bc_matrices-matrix.h5ad
filtered out 232 cells that have less than 200 genes expressed
filtered out 13246 genes that are detected in less than 3 cells
normalizing counts per cell
  finished (0:00:00)
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an
  AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
  with n_comps=30
  finished (0:00:00)
computing neighbors
  using 'X_pca' with n_pcs = 30
  finished: added to `.uns['neighbors']`
  `.obsp['distances']`, distances for each pair of neighbors
  `.obsp['connectivities']`, weighted adjacency matrix (0:00:23)
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 16 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:09)
ranking genes
  finished: added to `.uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:10)
scanpy==1.10.2 anndata==0.10.8 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph
==0.11.6 louvain==0.8.2 pynndescent==0.6.0
using 1 threads
wrote section timings to results/pbmc10k_section_times.csv
```

7. Console outputs (continued)

PBMC3K: cProfile run

```
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/louvain/__init__.py:54: UserWarning: pkg_resources is deprecated as an
API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as
2025-11-30. Refrain from using this package or pin to Setuptools<81.
from pkg_resources import get_distribution, DistributionNotFound
... reading from cache file cache/data-pbmc3k-filtered_gene_bc_matrices-matrix.h5ad
filtered out 19024 genes that are detected in less than 3 cells
normalizing counts per cell
finished (0:00:00)
extracting highly variable genes
finished (0:00:00)
--> added
'highly_variable', boolean vector (adata.var)
'means', float vector (adata.var)
'dispersions', float vector (adata.var)
'dispersions_norm', float vector (adata.var)
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an
AnnData. Making a copy.
view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
with n_comps=30
finished (0:00:00)
computing neighbors
using 'X_pca' with n_pcs = 30
finished: added to `uns['neighbors']`
`.obsp['distances']`, distances for each pair of neighbors
`.obsp['connectivities']`, weighted adjacency matrix (0:00:03)
running Louvain clustering
using the "louvain" package of Traag (2017)
finished: found 4 clusters and added
'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
finished: added
'X_umap', UMAP coordinates (adata.obsm) (0:00:05)
ranking genes
finished: added to `uns['rank_genes_groups']`
'names', sorted np.recarray to be indexed by group ids
'scores', sorted np.recarray to be indexed by group ids
'logfoldchanges', sorted np.recarray to be indexed by group ids
'pvals', sorted np.recarray to be indexed by group ids
'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:01)
scanpy==1.10.2 anndata==0.10.8 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph
==0.11.6 louvain==0.8.2 pynndescent==0.6.0
using 1 threads
wrote cProfile data to results/pbmc3k_rank_genes.prof
wrote cProfile summary to results/pbmc3k_rank_genes_profile.txt
```

7. Console outputs (continued)

PBMC6K: cProfile run

```
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/louvain/__init__.py:54: UserWarning: pkg_resources is deprecated as an
  API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as
  2025-11-30. Refrain from using this package or pin to Setuptools<81.
from pkg_resources import get_distribution, DistributionNotFound
... reading from cache file cache/data-pbmc6k-filtered_gene_bc_matrices-matrix.h5ad
filtered out 1 cells that have less than 200 genes expressed
filtered out 18395 genes that are detected in less than 3 cells
normalizing counts per cell
  finished (0:00:00)
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an
  AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
  with n_comps=30
  finished (0:00:00)
computing neighbors
  using 'X_pca' with n_pcs = 30
  finished: added to `uns['neighbors']`
  `obsp['distances']`, distances for each pair of neighbors
  `obsp['connectivities']`, weighted adjacency matrix (0:00:03)
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 7 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:10)
ranking genes
  finished: added to `uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:02)
scanpy==1.10.2 anndata==0.10.8 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph
==0.11.6 louvain==0.8.2 pynndescent==0.6.0
using 1 threads
wrote cProfile data to results/pbmc6k_rank_genes.prof
wrote cProfile summary to results/pbmc6k_rank_genes_profile.txt
```


7. Console outputs (continued)

PBMC10K: cProfile run

```
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/louvain/__init__.py:54: UserWarning: pkg_resources is deprecated as an
  API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as
  2025-11-30. Refrain from using this package or pin to Setuptools<81.
from pkg_resources import get_distribution, DistributionNotFound
... reading from cache file cache/data-pbmc10k-filtered_gene_bc_matrices-matrix.h5ad
filtered out 232 cells that have less than 200 genes expressed
filtered out 13246 genes that are detected in less than 3 cells
normalizing counts per cell
  finished (0:00:00)
extracting highly variable genes
  finished (0:00:00)
--> added
  'highly_variable', boolean vector (adata.var)
  'means', float vector (adata.var)
  'dispersions', float vector (adata.var)
  'dispersions_norm', float vector (adata.var)
/home/ysun564/venvs/bmi500-py312/lib/python3.12/site-packages/scanpy/preprocessing/_scale.py:316: UserWarning: Received a view of an
  AnnData. Making a copy.
  view_to_actual(adata)
... as `zero_center=True`, sparse input is densified and may lead to large memory consumption
computing PCA
  with n_comps=30
  finished (0:00:00)
computing neighbors
  using 'X_pca' with n_pcs = 30
  finished: added to `uns['neighbors']`
  `obsp['distances']`, distances for each pair of neighbors
  `obsp['connectivities']`, weighted adjacency matrix (0:00:23)
running Louvain clustering
  using the "louvain" package of Traag (2017)
  finished: found 16 clusters and added
  'louvain', the cluster labels (adata.obs, categorical) (0:00:00)
computing UMAP
  finished: added
  'X_umap', UMAP coordinates (adata.obsm) (0:00:09)
ranking genes
  finished: added to `uns['rank_genes_groups']`
  'names', sorted np.recarray to be indexed by group ids
  'scores', sorted np.recarray to be indexed by group ids
  'logfoldchanges', sorted np.recarray to be indexed by group ids
  'pvals', sorted np.recarray to be indexed by group ids
  'pvals_adj', sorted np.recarray to be indexed by group ids (0:00:14)
scanpy==1.10.2 anndata==0.10.8 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pandas==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph
==0.11.6 louvain==0.8.2 pynndescent==0.6.0
using 1 threads
wrote cProfile data to results/pbmc10k_rank_genes.prof
wrote cProfile summary to results/pbmc10k_rank_genes_profile.txt
```